

Ayan Kawlekar

Berkeley, CA | 609-934-2104 | ayan_kawlekar@berkeley.edu | [linkedin.com/in/ayan-kawlekar](https://www.linkedin.com/in/ayan-kawlekar) | github.com/AYANKAWLEKAR

EDUCATION

University of California, Berkeley

Berkeley, CA

B.A. Applied Mathematics, B.A. Data Science | Major GPA: 4.0, Cumulative GPA: 3.65

Expected Spring 2028

- **Coursework:** Differential Equations, Discrete Math, DECODE Startup Success, Data Structures

PROFESSIONAL EXPERIENCE

Oracle

Remote

Product Management Intern, Contract

June 2026 – Present

- Formulate demo agent using 26ai database; context-engineering through multi-hop traversal and 3 memory modalities
- Build in-database semantic retrieval with DBMS_VECTOR & ONNX MiniLM embeddings over 50,000+ career corpus
- Wrap tool call with observability functions to produce visual demo planned for over 50 sales calls using React,NextJs

Origo Capital

Remote

AI Intern

May 2026 – Present

- Develop Python-based AI agents automating sourcing and screening of 100+ LMM buyout targets (\$1–5M EBITDA)
- Support financial diligence on LMM buyout opportunities; drafted multiple CIMs and internal investment decks
- Maintain portfolio tracking for existing investments, updating key KPIs and summaries for internal reporting

ChipChop (Antler Backed)

San Francisco, CA

Product Development Intern

Feb. 2026 – Present

- Architect multi-agent workflow for FPGA design, transforming user requests and initial documents into RTL code
- Design proprietary debugging agent that assesses evidence and routes custom tests across agents; 70% bug coverage
- Manage shared agent memory through custom data store for workspace logs, action summaries, and user conversation
- Orchestrate containerized RTL simulations on Kubernetes, parallelizing test runs and parsing failure logs

Berkeley Nanotechnology Lab

Berkeley, CA

Machine Learning Research Assistant

Sep. 2025 – May 2026

- Designed PyTorch models to classify and process lithography manufacturing data and optimize machine parameters
- Built SEMI SECS/GEM extraction protocol, standardizing tool output and transforming data with Pandas
- Finetuned Llama-3 on 9,000+ lab samples for an internal agent that onboards members and queries past trials

Visa

Berkeley, CA

Software Engineer, Contract

Sep. 2025 – Dec. 2025

- Created AI web-scraping tools validating 1,600+ VCIS financial institutions against European regulators
- Wrote Playwright and Pydantic backend scripts standardizing regulator page content, using OpenAI to validate
- Implemented Dockerized ETL workflow ingesting Excel exports and generating reconciliation reports (JSON/XLSX)

Rutgers University PRACSYS Lab

New Brunswick, NJ

Robotics Research Assistant

June 2024 – Jan. 2025

- Deployed multimodal CV pipeline for industrial decision-making with RGB, depth, and segmentation on 340K+ scenes
- Trained and evaluated models on AWS SageMaker, running ablation studies on MultiMAE visual encoders
- Ran modality-testing studies, exporting via TorchScript to benchmark inference latency for ROS-based deployment

PROJECTS

StockSmart | Python, PyTorch, Stable-Baselines, Reinforcement Learning | github.com/AYANKAWLEKAR/RL

- Crafted retail restocking system pairing a Temporal Fusion Transformer forecaster with an RL ordering policy
- Trained DQN agent across 5 product categories with Stable-Baselines to time restock order placement

Private Equity Sourcing Engine | Python, DuckDB, pgvector | github.com/AYANKAWLEKAR/SourcingEngine

- Built tool-calling LLM agent and pgvector RAG retriever converting natural-language buy-boxes into filter rulesets
- Queried 4.4M corporate records in <15ms via DuckDB; built entity resolution reaching 95% ABN match accuracy
- Architected 3-tier scoring pipeline pairing hard-rule screening and statistical formulas with Claude moat judging

Memcache API | FastAPI, Celery, Redis, Neo4j | github.com/AYANKAWLEKAR/MemCache

- Constructed episodic memory infrastructure for long-running personal agents, improving context during retrieval
- Used Celery for summarization, Redis for conversation caching, Neo4j to graph entity and episode relations

TECHNICAL SKILLS

Languages: Python, SQL, C++, JavaScript, Java

Frameworks & Libraries: PyTorch, TensorFlow, NumPy, FastAPI, React, LangChain, Pandas, Pydantic, Celery

Developer Tools: AWS, Docker, Kubernetes, Git, Oracle 26ai, PostgreSQL, Redis, Neo4j, Tableau

Interests: Competition Robotics, Squash, Sports Analytics, Venture Capital, Poker, Coffee Brewing